

Chapter 8. Partial Derivatives and the Gradient

Coordinate rates of change, sensitivity, and local information

A function of one variable has one basic input direction: move left or move right on the number line. A function of several variables has many input directions. The first step is to measure change in coordinate directions.

For a function $f(x, y)$, the partial derivative f_x measures the instantaneous rate of change when x changes and y is held fixed. The partial derivative f_y measures the instantaneous rate of change when y changes and x is held fixed. Together they form the *gradient vector*

$$\nabla f(x, y) = \langle f_x(x, y), f_y(x, y) \rangle.$$

The gradient is the local direction-information vector for a scalar field. It tells us how the output changes to first order near a point, how level curves are oriented, and how a loss function changes when model parameters change.

Teaching-material connection

This chapter develops the material on partial derivatives, geometric traces, numerical derivative estimates, second partial derivatives, Clairaut's theorem, and the gradient vector. The main idea is that partial derivatives reduce multivariable differentiation to single-variable differentiation by holding all other variables fixed.

Learning goals

After studying this chapter, you should be able to:

1. define $f_x(a, b)$ and $f_y(a, b)$ using limits;
2. interpret partial derivatives as slopes of coordinate traces of a surface;
3. compute partial derivatives by treating other variables as constants;
4. estimate partial derivatives from data or finite differences;
5. compute second partial derivatives and mixed partial derivatives;
6. use Clairaut's theorem when mixed partial derivatives are continuous;
7. compute the gradient of functions on $\mathbb{R}^2$, $\mathbb{R}^3$, and $\mathbb{R}^n$;
8. explain why the gradient is perpendicular to level curves and level surfaces;
9. connect gradients with sensitivity analysis, image processing, statistics, and machine learning.

1 From one derivative to many partial derivatives

For a single-variable function $y = f(x)$, the derivative at $x = a$ is

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}.$$

Geometrically, this is the slope of the tangent line at $x = a$. For a two-variable function $z = f(x, y)$, the graph is a surface. At a point $(a, b, f(a, b))$, there are many possible curves on the surface passing through that point. The easiest ones are the two coordinate traces:

- the trace obtained by fixing $y = b$ and allowing x to vary;
- the trace obtained by fixing $x = a$ and allowing y to vary.

The slopes of these traces are the partial derivatives.

Core idea

A partial derivative is an ordinary derivative applied to a slice of a multivariable function.

2 Definitions of f_x and f_y

Let f be a function of two variables. The partial derivative of f with respect to x at (a, b) is

$$f_x(a, b) = \lim_{h \rightarrow 0} \frac{f(a + h, b) - f(a, b)}{h},$$

provided the limit exists. The partial derivative of f with respect to y at (a, b) is

$$f_y(a, b) = \lim_{h \rightarrow 0} \frac{f(a, b + h) - f(a, b)}{h},$$

provided the limit exists.

Equivalently, if $g(x) = f(x, b)$, then $f_x(a, b) = g'(a)$. If $k(y) = f(a, y)$, then $f_y(a, b) = k'(b)$.

The notation is flexible:

$$f_x = \frac{\partial f}{\partial x} = \frac{\partial z}{\partial x} = D_x f, \quad f_y = \frac{\partial f}{\partial y} = \frac{\partial z}{\partial y} = D_y f.$$

How to compute partial derivatives

To compute f_x , treat y as a constant and differentiate with respect to x . To compute f_y , treat x as a constant and differentiate with respect to y .

3 Geometric interpretation: slopes of traces

Suppose the surface is $z = f(x, y)$ and we focus on the point $(a, b, f(a, b))$. The trace in the plane $y = b$ is the single-variable curve $z = f(x, b)$. Its tangent slope at $x = a$ is $f_x(a, b)$. The trace in the plane $x = a$ is $z = f(a, y)$, and its tangent slope at $y = b$ is $f_y(a, b)$.

Example 3.1: A surface and its coordinate traces

Let

$$f(x, y) = 4 - \frac{x^2}{4} - y^2.$$

This is a downward-opening elliptic paraboloid. At the point $(1, 1)$,

$$f_x(x, y) = -\frac{x}{2}, \quad f_y(x, y) = -2y.$$

Thus

$$f_x(1, 1) = -\frac{1}{2}, \quad f_y(1, 1) = -2.$$

Interpretation:

- if $y = 1$ is held fixed and x increases near $x = 1$, the height decreases at rate $1/2$;
- if $x = 1$ is held fixed and y increases near $y = 1$, the height decreases at rate 2 .

Surface with coordinate traces

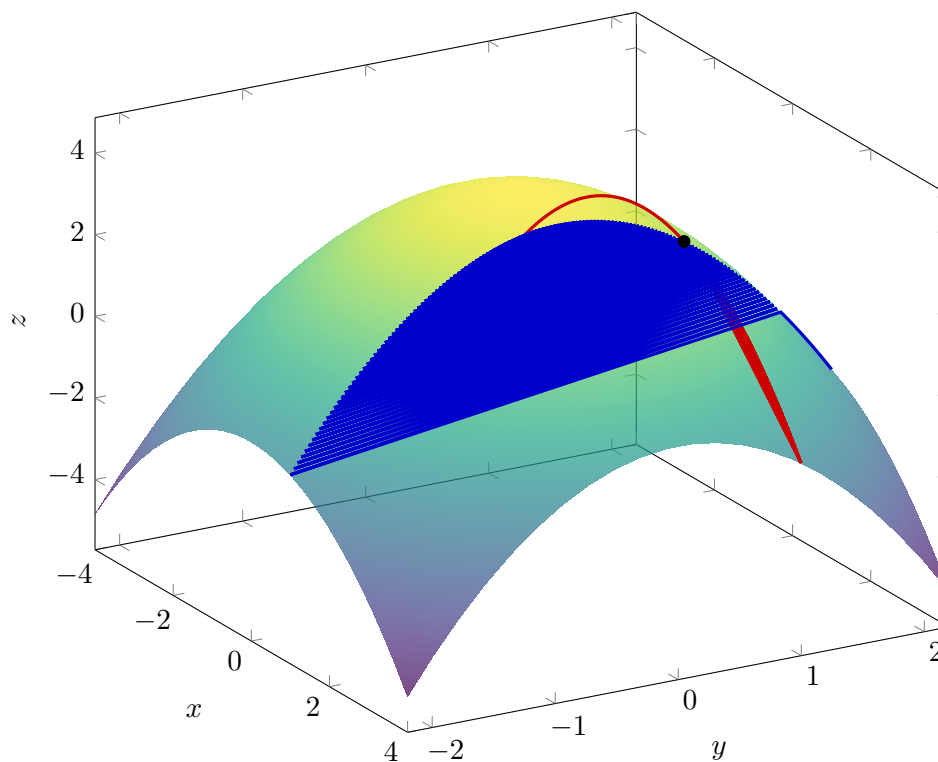


Figure 1: Coordinate traces of $f(x, y) = 4 - x^2/4 - y^2$ through the point $(1, 1, 2.75)$. The partial derivatives are the slopes of these traces.

4 Computing partial derivatives

Example 4.1: Polynomial, trigonometric, and exponential terms

Let

$$f(x, y) = x^3 + x^2 \sin y - xe^y.$$

To compute f_x , hold y constant:

$$f_x(x, y) = 3x^2 + 2x \sin y - e^y.$$

To compute f_y , hold x constant:

$$f_y(x, y) = x^2 \cos y - xe^y.$$

At $(1, 2)$,

$$f_x(1, 2) = 3 + 2 \sin 2 - e^2, \quad f_y(1, 2) = \cos 2 - e^2.$$

Example 4.2: A radial function

Let

$$f(x, y) = \sqrt{x^2 + y^2}.$$

For $(x, y) \neq (0, 0)$,

$$f_x(x, y) = \frac{x}{\sqrt{x^2 + y^2}}, \quad f_y(x, y) = \frac{y}{\sqrt{x^2 + y^2}}.$$

At $(1, 2)$,

$$f_x(1, 2) = \frac{1}{\sqrt{5}}, \quad f_y(1, 2) = \frac{2}{\sqrt{5}}.$$

At the origin, these formulas do not apply because the denominator vanishes. In fact, $f_x(0, 0)$ and $f_y(0, 0)$ exist, but the function is not differentiable there. Partial derivatives give coordinate-direction information, but not always the whole story.

5 Partial derivatives from data

In applications, we often do not know a symbolic formula for f . We may only have sampled values. Then partial derivatives are estimated by finite differences. For example,

$$f_x(1, 2) \approx \frac{f(1.1, 2) - f(1, 2)}{0.1}, \quad f_y(1, 2) \approx \frac{f(1, 2.15) - f(1, 2)}{0.15}.$$

This idea is fundamental in numerical differentiation, image processing, computer vision, and machine learning.

Example 5.1: Finite-difference estimates

For $f(x, y) = x^3 + x^2 \sin y - xe^y$, forward differences provide simple estimates for $f_x(1, 2)$ and $f_y(1, 2)$. A more accurate approximation often uses centered differences:

$$f_x(a, b) \approx \frac{f(a + h, b) - f(a - h, b)}{2h}.$$

The same idea works for f_y .

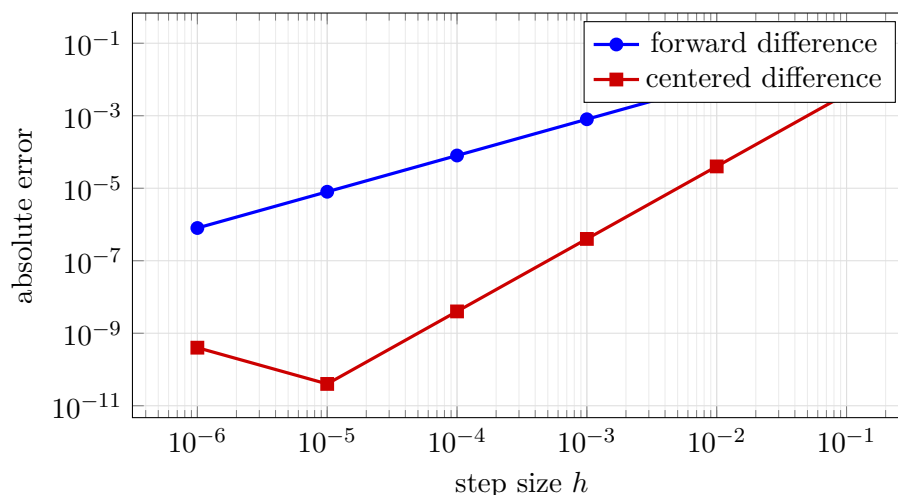


Figure 2: Centered differences usually converge faster than forward differences for smooth functions. The red curve drops more rapidly as h decreases.

6 Image processing interpretation

An image can be viewed as a function

$$I(x, y) = \text{pixel intensity at location } (x, y).$$

Then I_x measures horizontal change and I_y measures vertical change. Large values of $|I_x|$ or $|I_y|$ often indicate edges. This is the same calculus idea, but applied to a matrix of data instead of a symbolic formula.

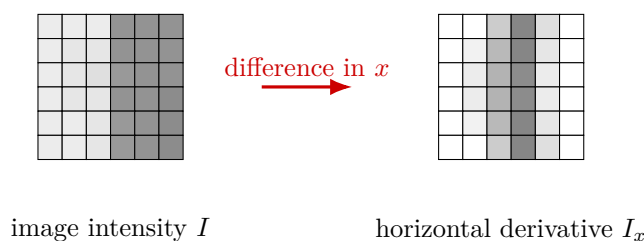


Figure 3: Finite differences applied to image-like data. Horizontal derivatives reveal sharp transitions, which are often edges.

Modern interpretation

In symbolic calculus, f_x and f_y are formulas. In data analysis, they are often approximations computed from samples, grids, or pixels. The concept is the same: measure local change in coordinate directions.

7 Second partial derivatives

If $z = f(x, y)$, then the four second partial derivatives are

$$\begin{aligned} f_{xx} &= \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right), & f_{xy} &= \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right), \\ f_{yx} &= \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right), & f_{yy} &= \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right). \end{aligned}$$

The derivatives f_{xy} and f_{yx} are the *mixed partial derivatives*.

Example 7.1: Second partials and mixed partials

Let

$$f(x, y) = x^2 + xy^3 + \sin(xy).$$

First derivatives:

$$f_x = 2x + y^3 + y \cos(xy), \quad f_y = 3xy^2 + x \cos(xy).$$

Second derivatives:

$$\begin{aligned} f_{xx} &= 2 - y^2 \sin(xy), \\ f_{xy} &= 3y^2 + \cos(xy) - xy \sin(xy), \\ f_{yx} &= 3y^2 + \cos(xy) - xy \sin(xy), \\ f_{yy} &= 6xy - x^2 \sin(xy). \end{aligned}$$

Here the mixed partial derivatives are equal.

8 Clairaut's theorem

Clairaut's theorem

Suppose f is defined on a disk containing (a, b) . If f_{xy} and f_{yx} are both continuous on that disk, then

$$f_{xy}(a, b) = f_{yx}(a, b).$$

In most elementary examples involving polynomials, exponentials, logarithms, and trigonometric functions on regions where the formulas are defined, the continuity hypothesis is satisfied. Then the order of mixed partial differentiation can be switched.

9 The gradient vector

The first partial derivatives form a vector. For $f(x, y)$,

$$\nabla f(x, y) = \langle f_x(x, y), f_y(x, y) \rangle.$$

For $f(x, y, z)$,

$$\nabla f(x, y, z) = \langle f_x(x, y, z), f_y(x, y, z), f_z(x, y, z) \rangle.$$

More generally, for $f(x_1, \dots, x_n)$,

$$\nabla f(x_1, \dots, x_n) = \left\langle \frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \dots, \frac{\partial f}{\partial x_n} \right\rangle.$$

The symbol ∇ is pronounced *del* or *nabla*.

Example 9.1: Gradient of a two-variable function

Let

$$f(x, y) = \sin(xy) + e^y.$$

Then

$$f_x = y \cos(xy), \quad f_y = x \cos(xy) + e^y.$$

Therefore

$$\nabla f(x, y) = \langle y \cos(xy), x \cos(xy) + e^y \rangle.$$

At $(0, 1)$,

$$\nabla f(0, 1) = \langle 1, e \rangle.$$

Example 9.2: Gradient of a three-variable function

Let

$$f(x, y, z) = xyz.$$

Then

$$f_x = yz, \quad f_y = xz, \quad f_z = xy.$$

Thus

$$\nabla f(x, y, z) = \langle yz, xz, xy \rangle.$$

At $(1, 2, 3)$,

$$\nabla f(1, 2, 3) = \langle 6, 3, 2 \rangle.$$

10 The gradient and level curves

Level curves have equations $f(x, y) = c$. The gradient ∇f is perpendicular to the level curve at points where $\nabla f \neq 0$.

Suppose $r(t) = \langle x(t), y(t) \rangle$ is a curve that stays on the level curve $f(x, y) = c$. Then $f(x(t), y(t)) = c$. Differentiating gives

$$\nabla f(x(t), y(t)) \cdot r'(t) = 0.$$

The vector $r'(t)$ is tangent to the level curve. Therefore the gradient is perpendicular to the tangent direction.

Geometric meaning of the gradient

The gradient points normal to level curves and level surfaces. It tells us which direction cuts most directly across the contours.

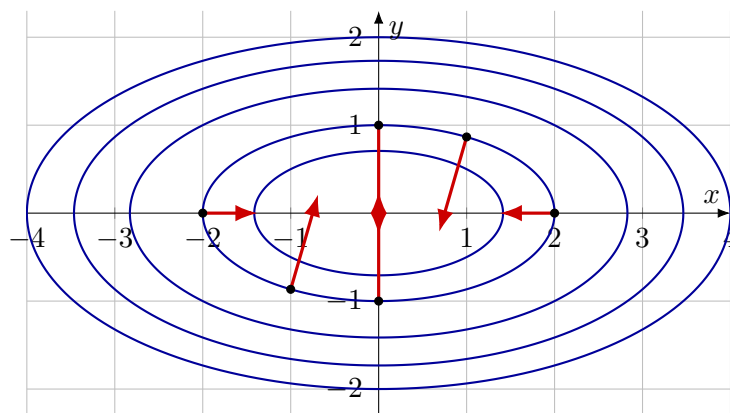


Figure 4: Gradient vectors are perpendicular to level curves of $H(x, y) = 4 - x^2/4 - y^2$.

11 Gradient fields

For each point (x, y) , the gradient produces a vector $\nabla f(x, y)$. The collection of these vectors is called a *gradient field*. Gradient fields show local directions of increase and form an important special class of vector fields.

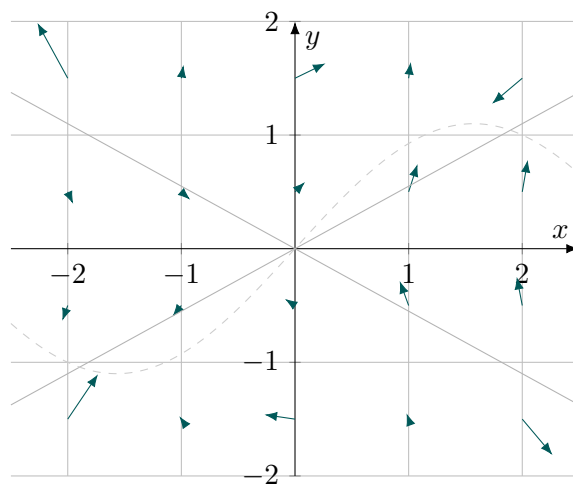


Figure 5: A gradient field for a smooth scalar function. The arrows encode local partial derivative information.

12 Directional derivatives as a preview

The next major idea is that we can measure rate of change in any unit direction u . If

$$u = \langle a, b \rangle$$

is a unit vector, the directional derivative is

$$D_u f(x, y) = \nabla f(x, y) \cdot u.$$

This formula says that the gradient contains all first-order directional information. The coordinate partial derivatives are special cases:

$$D_{\mathbf{i}}f = f_x, \quad D_{\mathbf{j}}f = f_y.$$

Thus the gradient is the vector that converts directions into rates of change.

Example 12.1: Directional derivative from a gradient

For

$$f(x, y) = \sin(xy) + e^y,$$

we found

$$\nabla f(0, 1) = \langle 1, e \rangle.$$

Let $v = \langle 2, 3 \rangle$. The corresponding unit vector is

$$u = \frac{v}{\|v\|} = \left\langle \frac{2}{\sqrt{13}}, \frac{3}{\sqrt{13}} \right\rangle.$$

Therefore

$$D_u f(0, 1) = \langle 1, e \rangle \cdot \left\langle \frac{2}{\sqrt{13}}, \frac{3}{\sqrt{13}} \right\rangle = \frac{2 + 3e}{\sqrt{13}}.$$

13 Gradients in statistics and machine learning

In statistics and machine learning, a model often has parameters

$$\theta = (\theta_1, \dots, \theta_n),$$

and we minimize a loss function $L(\theta_1, \dots, \theta_n)$. The partial derivative $\partial L / \partial \theta_k$ measures how sensitive the loss is to parameter θ_k while the other parameters are held fixed. The gradient $\nabla L(\theta)$ collects all parameter sensitivities into one vector.

For least-squares linear regression with data (x_i, y_i) and model $\hat{y}_i = \beta_0 + \beta_1 x_i$, the loss is

$$L(\beta_0, \beta_1) = \sum_{i=1}^m (y_i - (\beta_0 + \beta_1 x_i))^2.$$

The partial derivatives are

$$\frac{\partial L}{\partial \beta_0} = -2 \sum_{i=1}^m (y_i - (\beta_0 + \beta_1 x_i)),$$

and

$$\frac{\partial L}{\partial \beta_1} = -2 \sum_{i=1}^m x_i (y_i - (\beta_0 + \beta_1 x_i)).$$

These two numbers form the gradient in the parameter plane.

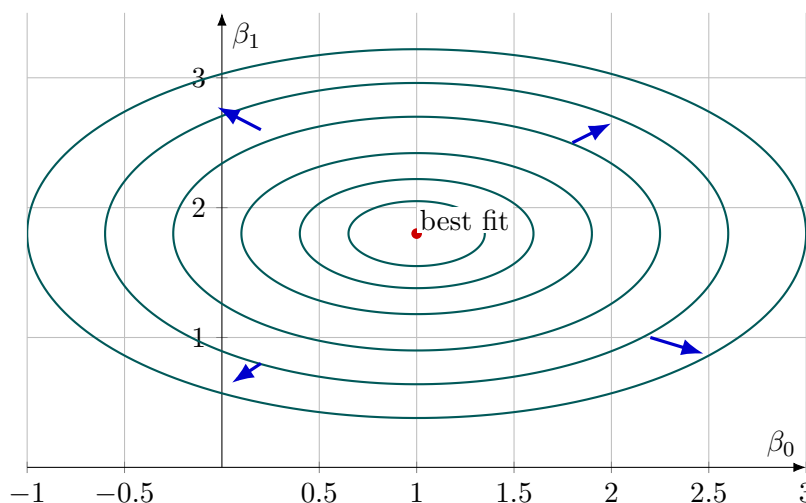


Figure 6: The gradient of a least-squares loss points toward increasing loss; the negative gradient points downhill.

14 Common mistakes

Mistake 1: Differentiating every variable at once

When computing f_x , only x varies. All other variables are constants. For example, if $f(x, y) = x^2y$, then $f_x = 2xy$, not $2xy + x^2$.

Mistake 2: Confusing partial derivatives with total change

The partial derivative $f_x(a, b)$ measures change only in the x direction. It does not describe motion along an arbitrary path unless the path moves only in the x direction.

Mistake 3: Forgetting to normalize a direction vector

In the directional derivative formula $D_u f = \nabla f \cdot u$, the vector u must be a unit vector. If a problem gives a nonunit direction vector, divide by its length first.

Mistake 4: Assuming mixed partials are always equal

Mixed partials are equal under hypotheses such as continuity near the point. For standard smooth examples this is usually true, but the theorem has a condition.

15 Chapter exercises

Conceptual questions

1. Explain in words what $f_x(a, b)$ means.

2. Explain why $f_x(a, b)$ is the slope of a trace of the surface $z = f(x, y)$.
3. Explain the difference between $f_x(a, b)$ and $f_y(a, b)$.
4. Why is a partial derivative not the same thing as a directional derivative in every direction?
5. Explain why ∇f is perpendicular to a level curve.
6. In an image, what do large values of $|I_x|$ and $|I_y|$ usually indicate?
7. In a loss function $L(\theta_1, \theta_2)$, what does $\partial L / \partial \theta_1$ mean?
8. Explain why the gradient is useful even when the function has many variables.

Skill problems

9. Find f_x and f_y for $f(x, y) = x^2y + 3xy^2 - y$.
10. Find f_x and f_y for $f(x, y) = e^{xy} + \ln(x + y)$ on its domain.
11. Find f_x and f_y for $f(x, y) = \cos(x^2 + y^2)$.
12. Find $\nabla f(1, 2)$ for $f(x, y) = x^3y + y^2$.
13. Find $\nabla f(0, 1)$ for $f(x, y) = \sin(xy) + e^y$.
14. Find $\nabla f(1, 2, 3)$ for $f(x, y, z) = x^2y + yz^3$.
15. Find all four second partial derivatives of $f(x, y) = x^2 + xy^3 + \sin(xy)$.
16. Verify Clairaut's theorem for $f(x, y) = e^{xy} + x^2y^3$.
17. Estimate $f_x(2, 1)$ using $f(2, 1) = 5.0$ and $f(2.1, 1) = 5.32$.
18. Estimate $f_y(2, 1)$ using $f(2, 1) = 5.0$ and $f(2, 1.2) = 4.7$.

Applied and computational problems

19. Let $H(x, y) = 1100 - 2x^2 - y^2$ model the height of a hill in meters. Compute H_x and H_y , and interpret them.
20. For the same hill, compute $\nabla H(10, 20)$ and explain what the signs of its components mean.
21. Let $T(x, y, z) = xe^y + 0.5z^2$. Compute $\nabla T(1, 0, 3)$.
22. Let I be an image intensity function. Explain how finite differences approximate I_x and I_y .
23. For least-squares regression with two parameters, derive the gradient of

$$L(\beta_0, \beta_1) = \sum_{i=1}^m (y_i - (\beta_0 + \beta_1 x_i))^2.$$

24. Write code to approximate $\nabla f(1, 2)$ for $f(x, y) = x^3 + x^2 \sin y - xe^y$ using centered differences.
25. Generate a contour plot of $f(x, y) = x^2 + 4y^2$ and draw gradient vectors at several points.
26. Compute numerical gradients on a grid for $f(x, y) = \sin x \cos y$.

Challenge problems

27. Let $f(x, y) = \sqrt{x^2 + y^2}$. Show that $f_x(0, 0)$ and $f_y(0, 0)$ exist, but explain why this does not mean the surface has a tangent plane at the origin.
28. Find a function for which f_x and f_y exist at the origin but the function is not continuous at the origin.
29. Suppose $f(x, y) = g(x^2 + y^2)$ for a differentiable single-variable function g . Find ∇f .
30. Suppose $f(x, y) = h(ax + by)$ for a differentiable single-variable function h . Find ∇f and describe the direction of the gradient.
31. Let $f(x, y, z) = F(x^2 + y^2 + z^2)$. Find ∇f and describe its direction geometrically.
32. In a high-dimensional model, the gradient has one component per parameter. Explain why this makes gradient methods scalable compared with searching every direction separately.

16 Python lab and interactive page

The Python lab for this chapter focuses on numerical and visual differentiation:

- estimating partial derivatives from function values;
- comparing forward, backward, and centered differences;
- plotting coordinate traces of surfaces;
- visualizing gradients on contour plots;
- computing image gradients;
- computing gradients of least-squares losses.

Resources:

- Chapter 8 Python lab: `../labs/Chapter08_Partial_Derivatives_and_the_Gradient.ipynb`
- Chapter 8 interactive HTML page: `../htmls/chapter-08.html`

17 Chapter summary

Partial derivatives are the coordinate-direction derivatives of a multivariable function. They are ordinary derivatives applied to slices of the function. For $f(x, y)$, the two first partial derivatives are f_x and f_y . For a function of n variables, there are n first partial derivatives.

The gradient packages these partial derivatives into a vector:

$$\nabla f = \left\langle \frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n} \right\rangle.$$

This vector is perpendicular to level sets and contains the first-order sensitivity information of the function. In modern applications, the gradient is the basic object behind numerical differentiation, image processing, optimization, maximum likelihood, backpropagation, and gradient-based learning.